Bangladesh University of Engineering and Technology

ME 352-Machine Design Sessional

Project Title: **Stress Analysis of a Bicycle Frame**

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Submitted by : Group No: B2-02

2010101 Mostafijur Rahaman

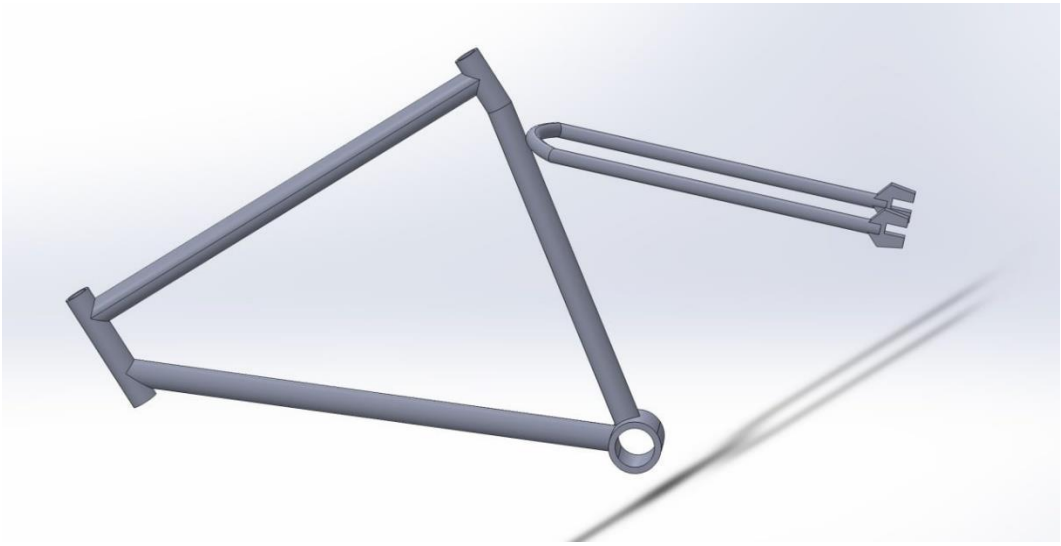
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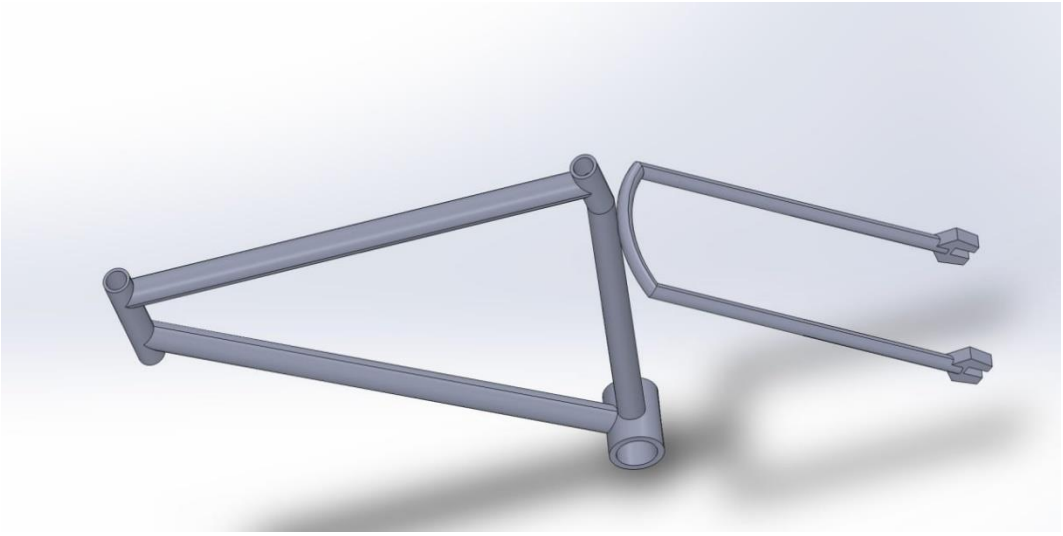
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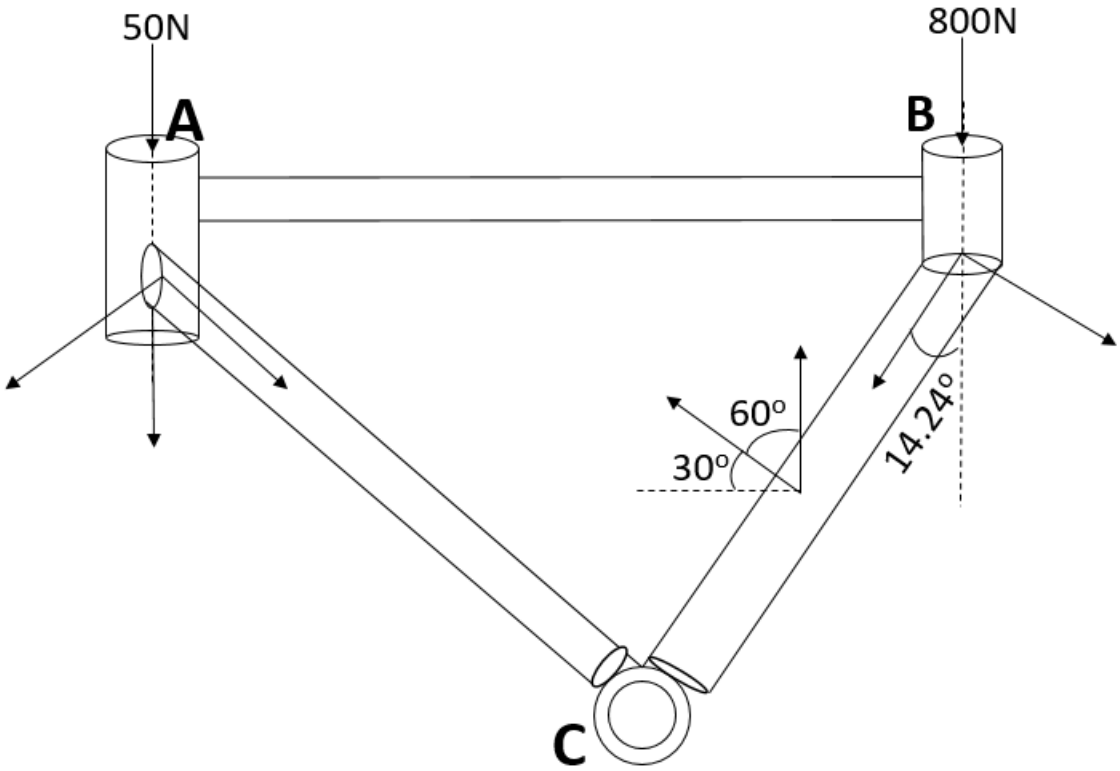
1810020 Monjurul Islam Jihad

SolidWorks Design





Hand Calculation



Stress Analysis at Point A:

$$\sum M_A = 50 \times 20 - 800 \times 723.64 = -577.92 \times 10^3 \, Nmm$$

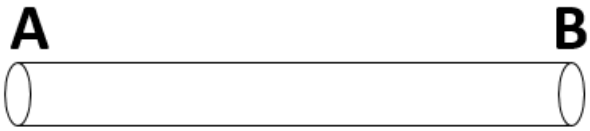
So, stress at A,

$$\sigma = \frac{F}{A} + \frac{MC}{I}$$

$$= \frac{50}{\pi(20^2 - 15.5^2)} + \frac{579.92 \times 10^3 \times 20}{\frac{\pi}{64}(40^4 - 31^4)}$$

$$= 36.07 MPa$$

Stress Analysis at Point B:



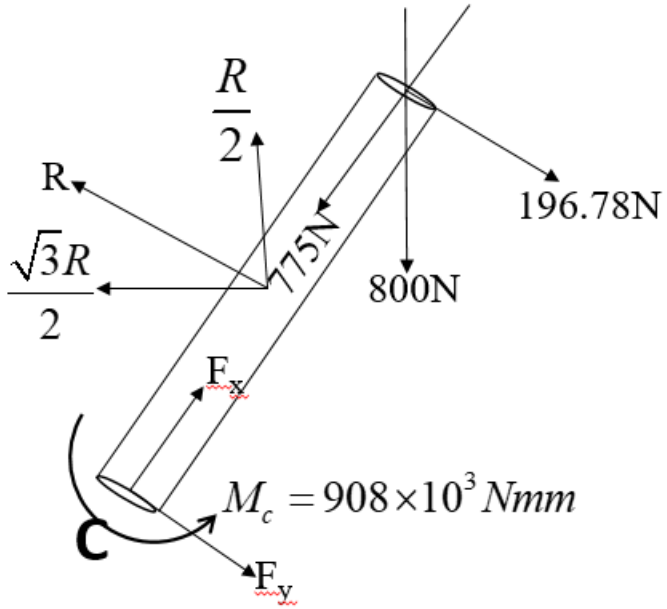
At point B,

$$\sigma = \frac{F}{A} + \frac{MC}{I}$$

$$= \frac{800}{\pi\left(20^2 - 15.5^2\right)} + \frac{20.282\times10^3\times10}{\frac{\pi}{4}\left(40^4 - 31^4\right)}$$

$$= 4.1MPa$$

Stress Analysis at Point: C



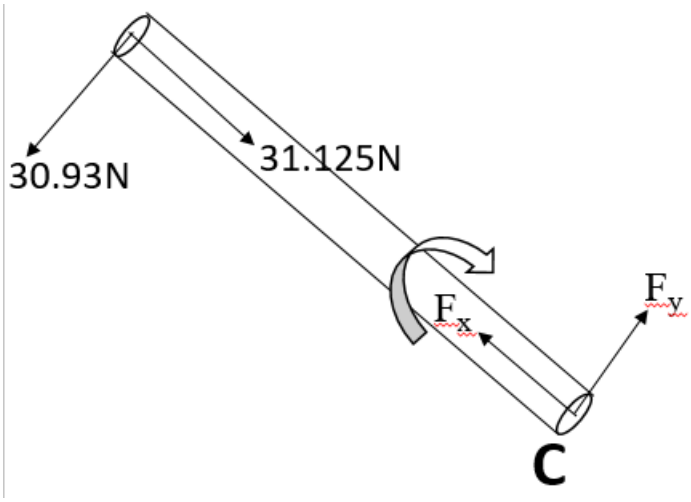
F_x=775N

R-F_y=196.78N

$$M_c = R\times453.37 - 196.78\times513 = 908\times10^3\ Nmm$$

$$\therefore R = 2226.6N$$

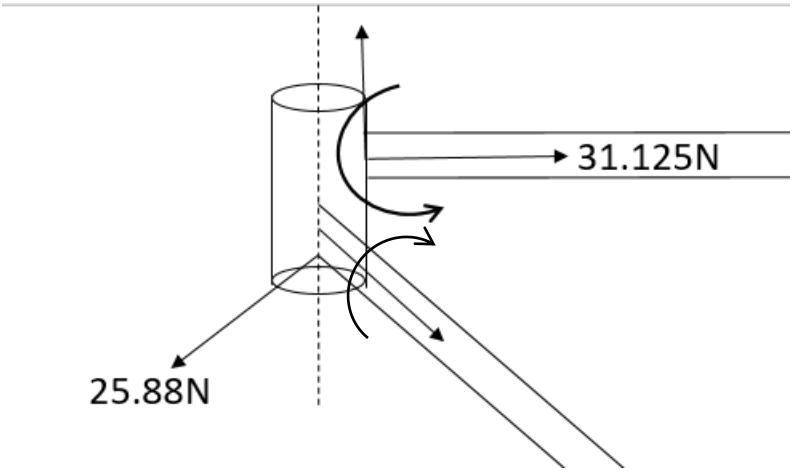
$$F_y = 25.88N$$

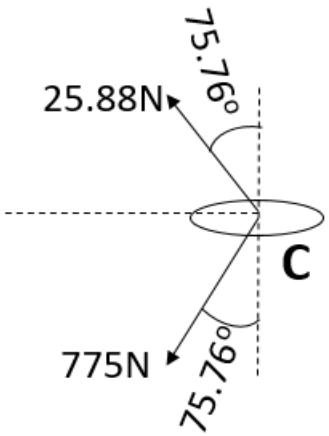


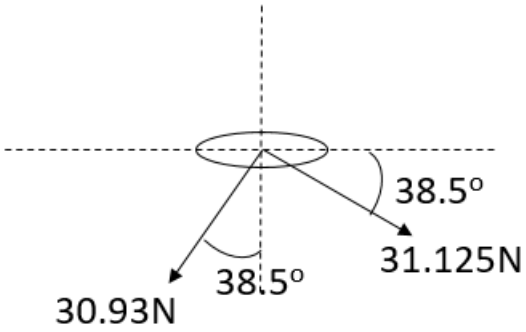
$F_x = 31.125N$

$F_y = 30.93N$

$M_o = 30.93 \times 740.98 = 22918.51Nmm$







$F_{y1} = 744.82N \downarrow$

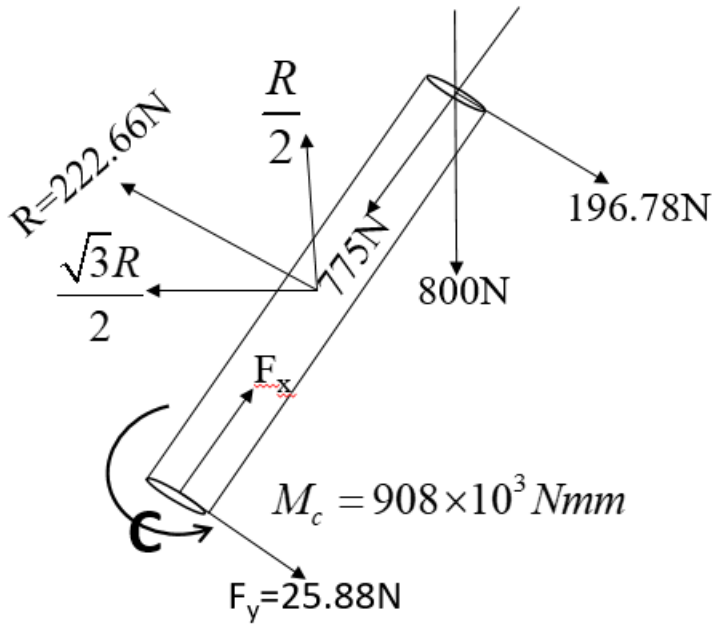
$F_{y2} = 43.38N \downarrow$

$$\sigma = \frac{F_{y1} + F_{y2}}{A} = 158MPa$$

$$\tau = \frac{Tr}{J}$$

$$= \frac{22918.51 \times 40.03}{\frac{\pi}{32}\left(80^4 - 60^4\right)}$$

$$= 0.33MPa$$



Moment at D,

$$M_D = 25.88 \times 453 - 196.78 \times 60 = -83.16 Nmm$$

$$\sigma = \frac{MC}{I} = \frac{83.16 \times 20}{\frac{\pi}{64}\left(40^2 - 30^2\right)} = 48.4 MPa$$

Axial Stress,

$$\sigma_A = \frac{775}{\pi\left(20^2 - 15^2\right)} = 1.4 MPa$$

Radial Stress,

$$\sigma_R = \frac{2226.6}{\pi\left(8^2 - 7^2\right)} = 47.25 MPa$$

$$\therefore \sigma = \sqrt{\left(44.8\right)^2 + \left(47.25\right)^2} = 68 PMPa$$

Safety Factor,

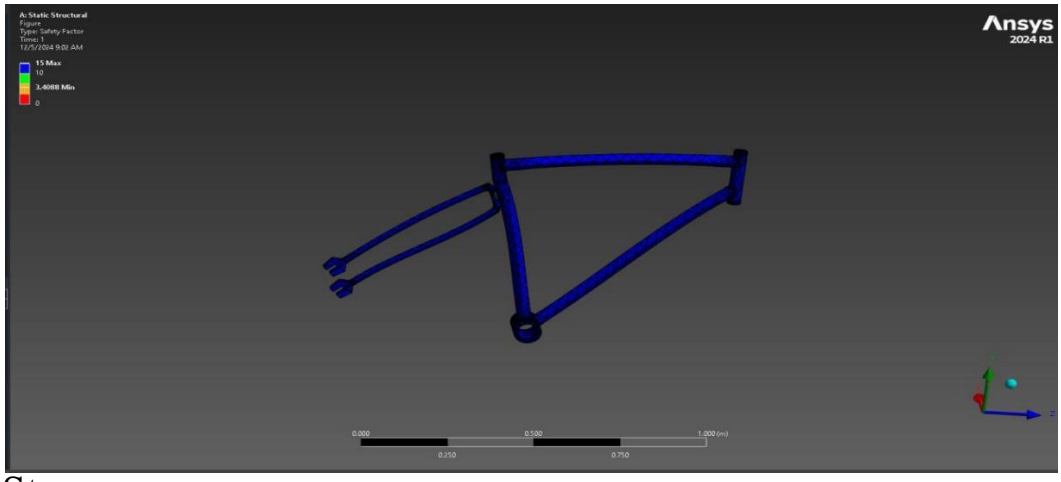
$$n = \frac{S_y}{\sigma}$$

Here, S_Y for steel=250MPa

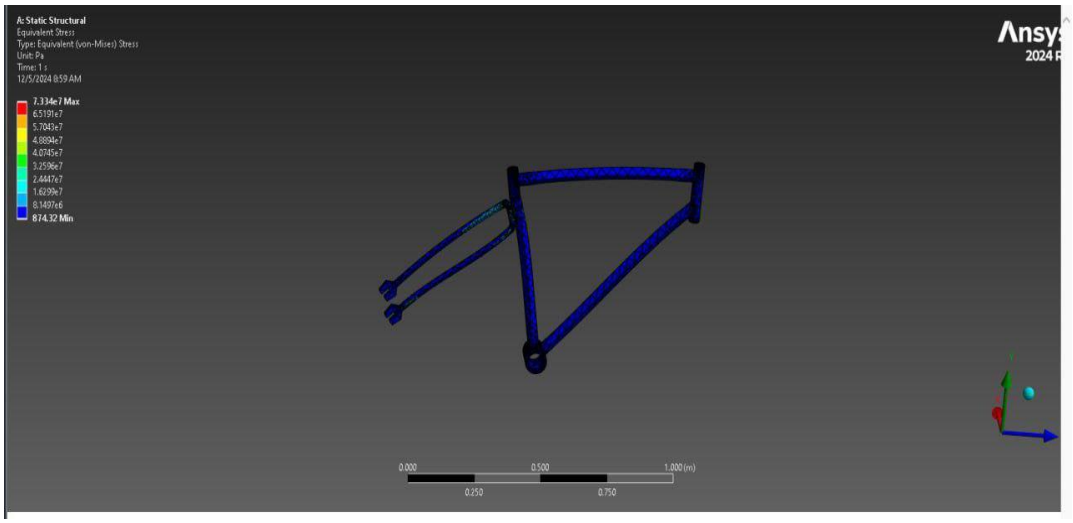
$$\therefore n = \frac{250}{68} = 3.67$$

Ansys Simulation

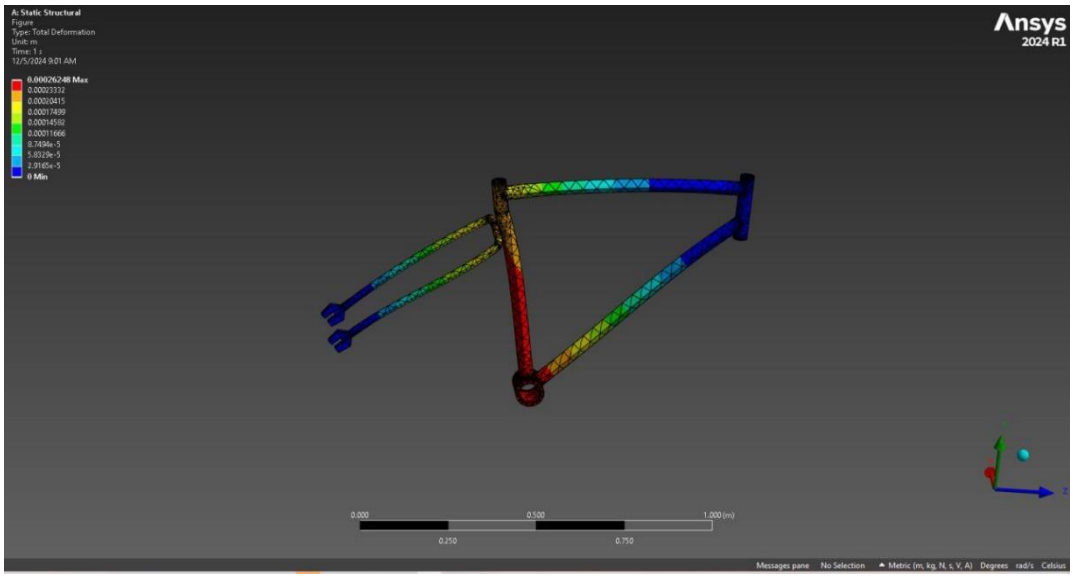
Mesh:



Stress:

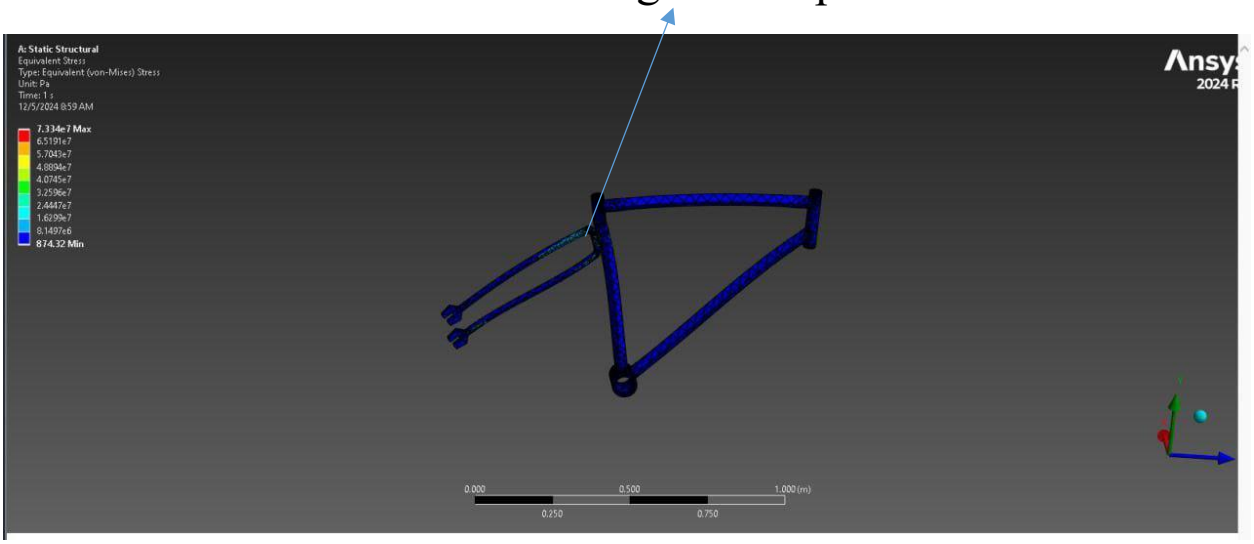


Strain:



Description of Ansys Simulation

Stress: high stress point

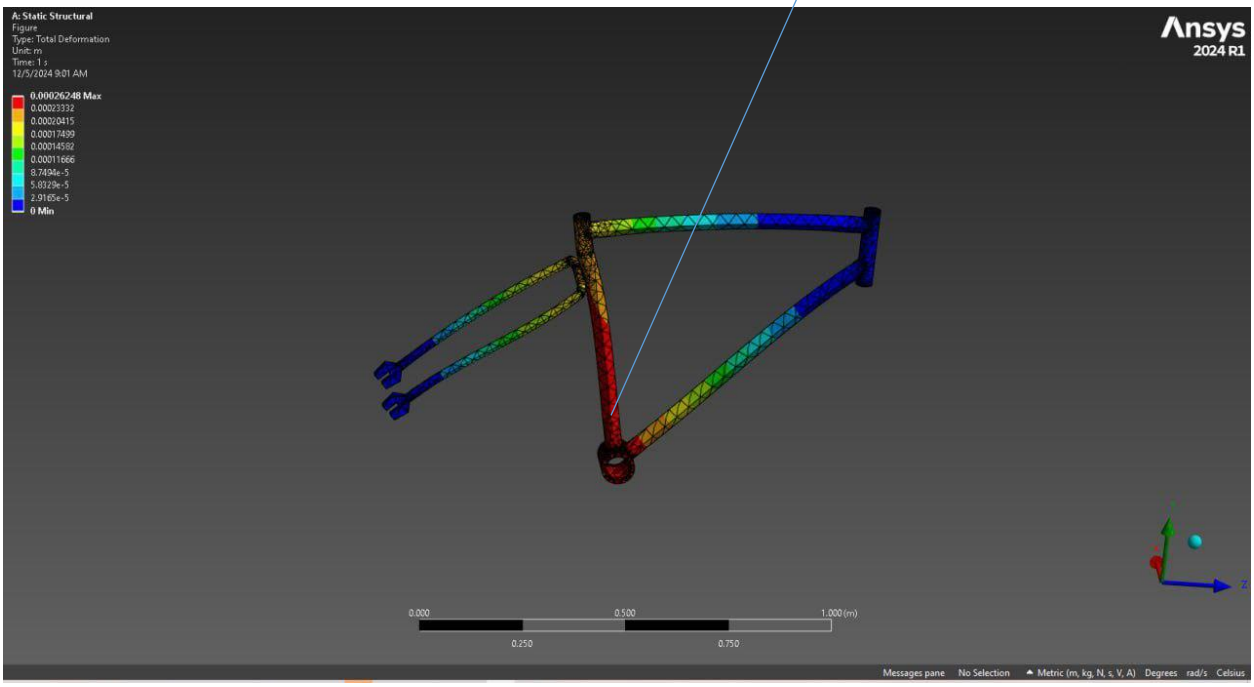


This is a **static structural stress analysis** of a bicycle frame, visualized in ANSYS. This image show's equivalent von Mises stress distribution along the bicycle frame. Hear, high stress zone are red zone and comparatively low stress zone are blue zone.

From our image we see that critical point or high stress point is the joining point of the triangular main frame and rear stays. Hear maximum stress is 73 MPa. Other part of the frame remains at low stress since it indicates the color of the rest part will be blue.

Since, the **most vulnerable point** is the joining point of triangular main frame and rear stays , that's why it requires design modification or material reinforcement and high factor of safety for this point. **Other part** of the bicycle indicate low stress regions(in blue) indicates less critical areas that could potentially be optimized to save material or weight.

Strain: high strain point



This is a **static structural strain analysis** of a bicycle frame, visualized in ANSYS. This image show's the total deformation along the bicycle frame. Hear, high strain zone are red zone and comparatively low strain zone are blue zone.

Maximum displacement occur where loads are applied and where the structure is less rigid. It also occur at the joint where forces are concentrated.

Hear, red region show's maximum deformation, blue region show's low deformation and yellow regions are the transition region. From our image we see that , red region are at the **lower section of the main triangle** and **joints where forces are concentrated.** That's why it represents the maximum deformation. From ANSYS analysis we see that this deformation is about 0.26248 mm. Blue regions are more rigid, that's why experience minimal displacement, often due to structural support

or boundary conditions. Care must be taken on the high strain region so that it can withstand at this strain condition and use ductile material instead of brittle material. Lower strain region can be optimized by considering cost and weight of the material.

Conclusion:

1. Most critical point of the bicycle frame on the basis of maximum von mises stress is the joining point of the triangular main frame and rear stays.
2. Most critical point of the bicycle frame on the basis of total deformation of the frame is the lower section of the main triangle.
3. Additional design consideration and care must be taken at this points, so that the frame can withstand at adverse condition.